# TAPASH KUMAR

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## SUMMARY

Data Scientist specializing in Machine Learning and Deep Learning to solve complex analytical challenges. Proficient in building predictive models with TensorFlow and Keras, performing advanced data modelling with SQL, and delivering actionable business insights through Power BI. Experienced in deploying end-to-end AI solutions that combine technical rigor with practical scalability to drive informed decision-making.

## TECHNICAL SKILLS

- Programming & Databases: Python, Microsoft Server SQL, PostgreSQL
- Data Analysis & Statistics: Data Cleaning, Exploratory Data Analysis (EDA), Statistical Analysis, Feature Engineering
- Machine Learning: Supervised and Unsupervised Learning, Hyperparameter Tuning, Model Evaluation
- Deep Learning: Convolutional Neural Network (CNN), Recurrent Neural Networks (RNN), Natural Language Processing (NLP)
- Libraries & Frameworks: NumPy, Pandas, scikit-learn, TensorFlow, Keras
- Generative AI: Large Language Models (LLMs), LangChain, Hugging Face Transformers, Retrieval-Augmented Generation (RAG)
- Vector Search & Retrieval: Vector Embeddings, FAISS, Chroma, Pinecone, Semantic Search
- Visualization & BI Tools: Matplotlib, Seaborn, Power BI, Excel
- Deployment, MLOps & Cloud: Flask, FastAPI, Streamlit, Docker, CI/CD, Git/GitHub, Jupyter, Microsoft Azure

## EXPERIENCE

**Stabor India Private Limited - AI Engineering Intern**

Gurugram Dec 2025 – Present

- Developed and optimized an **XGBoost model** achieving **15-20% improvement** in demand forecast accuracy on real-time datasets.
- Architected high throughput data pipelines to process continuous time series streams for instantaneous model consumption.
- Tuned model hyperparameters including L1/L2 regularization to effectively mitigate overfitting during training.

**CSIR – National Physical Laboratory (NPL) - Intern**

New Delhi Jun 2025 – Jul 2025

- Implemented **custom Allan Deviation** algorithm to precisely characterize short-term and long-term noise processes in atomic clocks.
- Estimated atomic clock frequency drift and successfully benchmarked performance results against global BIPM reference standards.
- Realized **20% alignment** with BIPM standards, significantly enhancing overall confidence in long-term stability assessments.

**ISRO Telemetry, Tracking and Command Network (ISTRAC) - Intern**

Bangalore Jun 2024 – Jul 2024

- Refined a microwave cavity design in Ansys to target a specific **TE011 mode** frequency of **1.42040575177 GHz**.
- Attained high precise frequency alignment with a minimal **43.99 Hz deviation**, significantly boosting Hydrogen Maser stability.
- Optimized cavity Q-factor performance through material property exploration, enhancing signal quality for precise timing.

## PROJECTS

**NLP-Based Movie Recommendation System** Demo

Tech Stack: LangChain, FAISS, HuggingFace, FastAPI, PostgreSQL, Docker, Next.js, Firebase Auth

- Built **FastAPI** recommender generating **HuggingFace embeddings via LangChain** and retrieving similar movies using FAISS indexing.
- Processed **5K+ movie** metadata vectors enabling semantic similarity search across plots, genres, and keywords.
- Achieved sub-second recommendations by optimizing **FAISS vector index** and embedding inference pipeline.
- Deployed full-stack system with **Next.js, PostgreSQL, Firebase Auth, and Docker** enabling scalable personalized recommendations.

**Brain MRI Prediction Using CNN**

Tech Stack: Python, Scikit-learn, Tensorflow, Keras, Matplotlib, Seaborn

- Engineered a custom **TensorFlow/Keras CNN** for brain tumor detection using data augmentation and hyperparameter tuning.
- Increased **model accuracy by 12%** over the baseline through rigorous image preprocessing and architectural refinement.
- Evaluated model performance metrics using scikit-learn, maintaining a **94% F1-score** to balance sensitivity and specificity.

**AI Intelligent Chatbot**

Tech Stack: LangGraph, LangChain, Streamlit, SQLite

- Built stateful conversational chatbot using LangGraph checkpoints to maintain persistent memory across sessions.
- Implemented LangChain orchestration for prompt management and automated conversation titling.
- Developed Streamlit interface with SQLite persistence for chat history storage and session continuity.

## CERTIFICATES

- Google Data Analytics Professional Certificate
- IBM AI Enterprise Workflow
- Pregrad Machine Learning Course
- IIRS AI/ML for Geodata Analysis

## EDUCATION

Delhi Technological University (DTU) | **B.Tech. Engineering Physics** | Oct 2022 – May 2026

## ACHIEVEMENTS

- Published research paper **"Innovative Cavity Design for Hydrogen Masers Using Simulation Methods"** in ISJEM (Mar 2026).